

Ajoy Dey

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Research Interests: Cybersecurity, Machine Learning, Natural Language Processing

EDUCATION

Bangladesh University of Engineering and Technology

B.Sc. in Computer Science and Engineering; CGPA 3.89/4.0

Dhaka, Bangladesh

Feb. 2020 – March 2025

Chattogram College

HSC; GPA 5.0/5.0

Chittagong, Bangladesh

2019

Uttar Kattali Al-Haj Mostafa Hakim High School

SSC; GPA 5.0/5.0

Chittagong, Bangladesh

2017

EXPERIENCE

Software Engineer I – Pathao

Full-time

Bangladesh

May 2025 – Present

- Implemented a new routing algorithm in a backend ride service to provide highly reliable ride routes, improving route reliability to 90% for over 100,000 rides completed daily.
- Developed and deployed an MLOps pipeline for ride ETA (pickup-to-dropoff) prediction using live timestamp datasets, achieving 85% prediction accuracy in production.
- Worked on backend system optimization and large-scale ride data processing to improve service efficiency and scalability.

TECHNICAL SKILLS

Language: Golang, Java, Python, JavaFX, C, C++, Bash, LATEX, SQL, Assembly

Web Technologies: HTML, CSS, JavaScript, React.js, Node.js

Library: Sklearn, Pandas, Matplotlib, Seaborn

Database Systems: Oracle, MongoDB

Other Tools: IntelliJ IDEA, Visual Studio Code, Docker, Wireshark

THESIS

Sentence-level Bangla Spell Error Corrector

My Undergraduate Thesis is supervised by Dr. Muhammad Abdullah Adnan sir

GitHub

- Developed a seq2seq transformer model ("MarianMT") fine-tuned on 3.8M sentence pairs generated using the "Synthetic Error Dataset Generation Mimicking Bengali Writing Pattern" algorithm.
- Utilized pre-trained checkpoints from Bangla Grammatical Error Correction (BGECC) tasks for knowledge transfer.
- Achieved 93% character-level accuracy, exploring Knowledge Distillation to improve performance further.

MACHINE LEARNING PROJECTS

Projects

All codes are written in Python

GitHub

- Project 01: Data Preprocessing and Feature Engineering
- Project 02: Logistic Regression with Bagging and Stacking
- Project 03: Function Approximation with Neural Network and Backpropagation
- Project 04: Principal component analysis (PCA) and Expectation-Maximization (EM) algorithm

Handwritten-Bangla-Word-Recognition

It is a machine learning project using EfficientNet-B0 model fine-tuned with the Zilla-64 dataset.

GitHub

- Achieved testing accuracy of 93.3%

SOFTWARE PROJECTS

Food Bear

This website built on the MERN stack which allows users to order food from various restaurants conveniently. [Github](#)

- Developed a food delivery website supporting restaurant search, order tracking, cart management, and home kitchen functionality.
- Integrated real-time features like order tracking, restaurant analytics dashboard, and voucher-based discounts.
- Implemented Google Maps integration for location-based restaurant search.

OTHER PROJECTS

Artificial Intelligence Offlines

[Github](#)

- Project 01: Solve n-Puzzle
- Project 02: Adversarial Search
- Project 03: Solving the Max-cut problem by greedy randomized adaptive search procedure
- Project 04: Decision tree learning for car evaluation

Computer Security Offlines

[Github](#)

- Project 01: Implemented a cryptosystem that uses a symmetric key for encryption and decryption. The symmetric key is securely shared by the Elliptic Curve Diffie Hellman key exchange method.
- Project 02: Cross-Site Scripting (XSS) Attack
- Project 03: Buffer Overflow Attack

Computer Graphics Offlines

[Github](#)

- Project 01: Assignment on OpenGL
- Project 02: Develop the raster-based graphics pipeline used in OpenGL.
- Project 03: Generated realistic images for a few geometric shapes using ray tracing with appropriate illumination techniques.

Operating Systems Offlines

[Github](#)

- Project 01: Shell Scripting Assignment
- Project 02: Xv6 - System Call
- Project 03: Xv6 – Scheduling
- Project 04: IPC Assignment

Football Player Database System

[Github](#)

It is built on Java, JavaFX with a server-client architecture, allowing clients to view and buy players.

- The server reads player data from a file, handles client connections, and updates player lists based on client interactions.

ACHIEVEMENTS

- **Merit List (BUET, 2023)** – Secured a place in the BUET Merit List.
- **Dean's List (BUET, 2021, 2022)** – Recognized for outstanding academic performance.
- **Board General Scholarship (HSC, 2019)**
- **Board General Scholarship (SSC, 2017)**

REFERENCES

Dr. Muhammad Abdullah Adnan

Professor

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Dr. Mohammad Saifur Rahman

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